Σ	the target formalism
$\mathcal{X}$	natural-language strings
$\mathcal{Y}_\Sigma$	well-formed formulas of Σ
$T_1 : \mathcal{X} \to \mathcal{Y}_\Sigma$	formalize
$T_2 : \mathcal{Y}_\Sigma \to \mathcal{X}$	back to natural language
$T_3 : \mathcal{X} \to \mathcal{Y}_\Sigma$	formalize again
check	$T_1(x) \;\equiv_\Sigma\; T_3\big(T_2(T_1(x))\big)$
as a solver query	`(assert (not (= y_orig y_rt)))` `(check-sat)` with $y^{\text{orig}} = T_1(x)$ and $y^{\text{rt}} = T_3(T_2(T_1(x)))$